

Question_8:

Code:

```
1  /*
2  Question No: 8
3  Maintain employee salaries while processing two types of queries.
4  ! k x -> change employee k's salary to x
5  ? a b -> count salaries in the range [a,b]
6  Fenwick Tree + coordinate compression is used.
7  */
8
9  #include <bits/stdc++.h>
10 using namespace std;
11
12 vector<int> bit;
13 int compare(const void *a,const void *b) {
14     return *(int*)a-*(int*)b;
15 }
16 void update(int i,int n,int x) {
17     while(i<=n) {
18         bit[i]+=x;
19         i+=i&-i;
20     }
21 }
22 int sum(int i) {
23     int s=0;
24
25     while(i) {
26         s+=bit[i];
27         i-=i&-i;
28     }
29     return s;
30 }
31 int main() {
32     int n,q;
33     cin>>n>>q;
34
35     vector<int>a(n+1),v;
36     for(int i=1;i<=n;i++) {
37         cin>>a[i];
38         v.push_back(a[i]);
```

```
39     }
40     vector<tuple<char,int,int>> qu(q);
41
42     for(auto &[c,x,y]:qu) {
43         cin>>c>>x>>y;
44         if(c=='!')
45             v.push_back(y);
46     }
47     sort(v.begin(),v.end());
48     v.erase(unique(v.begin(),v.end()),v.end());
49
50     bit.assign(v.size()+1,0);
51     for(int i=1;i<=n;i++) {
52         int p=lower_bound(v.begin(),v.end(),a[i])-v.begin()+1;
53         update(p,v.size(),1);
54     }
55     for(auto [c,x,y]:qu) {
56         if(c=='!') {
57             int old=lower_bound(v.begin(),v.end(),a[x])-v.begin()+1;
58             int nw=lower_bound(v.begin(),v.end(),y)-v.begin()+1;
59
60             update(old,v.size(),-1);
61             update(nw,v.size(),1);
62             a[x]=y;
63         }
64         else {
65             int l=lower_bound(v.begin(),v.end(),x)-v.begin();
66             int r=upper_bound(v.begin(),v.end(),y)-v.begin();
67             cout<<sum(r)-sum(l)<<"\n";
68         }
69     }
70 }
```

Output:

Test Case_1:

```
5 3
3 7 2 2 5
? 2 3
! 3 6
? 2 3
3
2
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
5 3
31 17 12 12 15
? 12 31
! 3 6
? 2 13
5
2
PS C:\Users\Prabin Yadav\
```